

Energy and Water Management for Industrial Large-Scale Water Networks: A Systematic Simultaneous Optimization Approach

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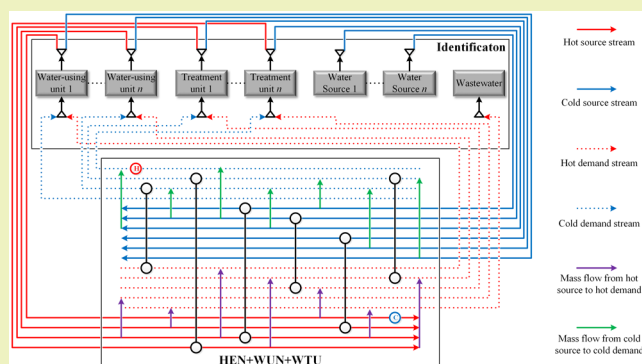
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Supporting Information

ABSTRACT: Water and energy management issues in process industries are generally related to the synthesis of heat integrated water networks (HIWN), because water and energy are inextricably intertwined in process water networks. The aim of this study is to present a novel mathematical programming model for HIWN synthesis problems and develop an efficient solution strategy. This research is an extension of our previous work, where the targeting of heat integrated water-using networks (HIWUN) has been addressed. In this model, wastewater treatment units and multiple contaminants are embedded, and the total annual cost (TAC) is optimized rather than targeting a simplified TAC. What is more, a three-step solution strategy is proposed to guarantee obtaining promising solutions. The freshwater consumption, the relaxed TAC (rTAC), and the TAC are optimized successively. Good initial points for the rigorous HIWN model in the last step can be generated by the minimizing the rTAC. Four examples including two large-scale examples are illustrated to demonstrate the applicability of the proposed model and the efficiency of the solution strategy. The results indicate that the proposed model is suitable for the synthesis of HIWN, and more significantly, better results for large-scale problems are achieved.

KEYWORDS: Heat integrated water networks, Multiple contaminants, Wastewater treatment, Solution strategy, MINLP model



INTRODUCTION

Sustainable development is one of most significant challenges facing society. The modern concept of sustainable development is derived mostly from the 1987 Brundtland Report,¹ in which it means “meeting the needs of the present without compromising the ability of future generations to meet their own needs”. Sustainability is essential for process industries, because they consume large amounts of natural resources, such as water² and energy,³ and cause waste discharge problems, such as wastewater.⁴ To make the process industry sustainable, both of its resource consumption and waste discharge should be significantly constrained. In the past decades, the process integration method has been recognized as a systematical and effective method for pollution reduction and energy saving. It is especially proficient in generating efficient strategies for water and energy management in chemical process industries. Several comprehensive reviews about the synthesis of water networks (WN) can be found in Bagajewicz,⁵ Foo,⁶ and Jezowski.⁷ Additionally, the reader can refer to Furman and Sahinidis⁸ for a comprehensive review of the synthesis of heat exchanger networks (HEN). However, water and energy are inextricably intertwined inside process water networks. Energy is consumed in process water networks for various heating or cooling purposes. This indicates

that water and energy management should be considered systematically and simultaneously. A state-of-art review about the synthesis of heat integrated water networks (HIWN) has been recently contributed by Ahmetović et al.⁹

The systematic methods for the synthesis of HIWN can be classified into two categories: sequential and simultaneous methods. In the sequential methods, the overall problem is decomposed into several subproblems, such as WN and HEN. The first work was contributed by the research group in the University of Manchester^{10–13} using conceptual design methods, which are sequential in nature. A two-stage procedure was developed to simultaneously minimize water and energy in two different systems: no water reuse¹² and maximum water reuse.¹³ A two-dimensional grid diagram and separate system approach were introduced for the synthesis of water-using networks (WUN) and HEN, respectively. Subsequently, many insightful conceptual design approaches have been proposed, such as Water Energy Balance Diagram,¹⁴ Heat Surplus Diagram,¹⁵ Superimposed Mass and Energy Curves (SMEC),¹⁶ and

Received: October 15, 2017

Revised: December 4, 2017

Published: December 13, 2017



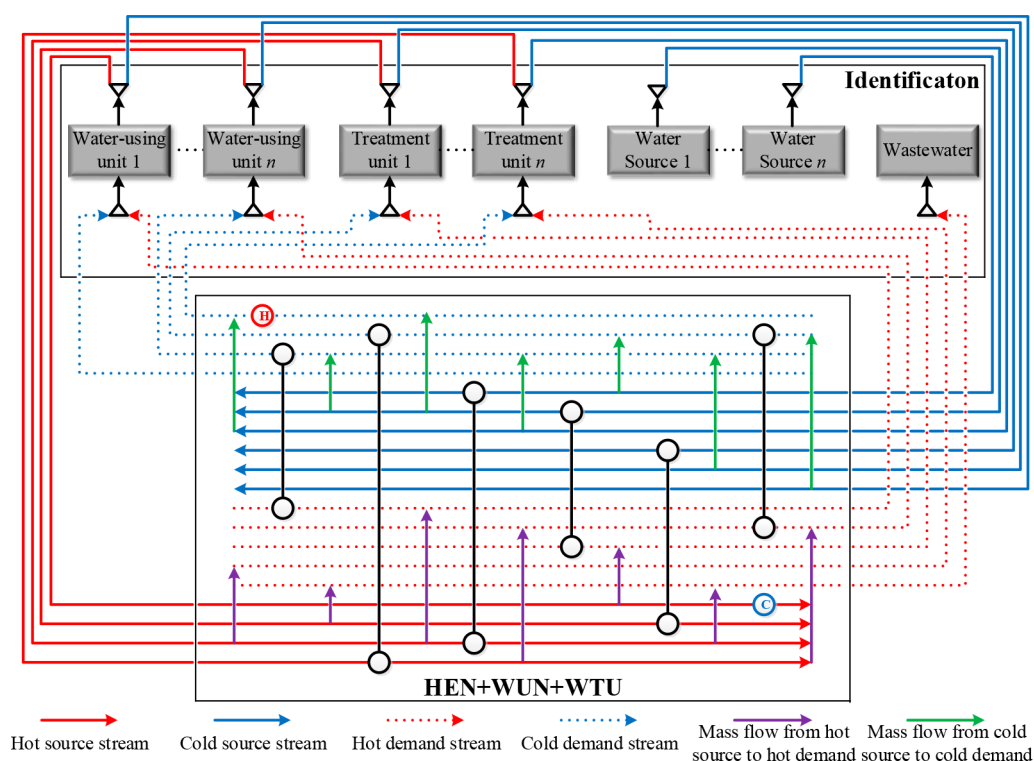


Figure 1. Representation of HIWN.

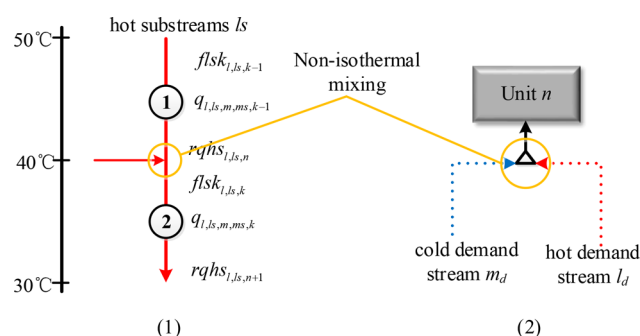


Figure 2. Illustration for nonisothermal mixing.

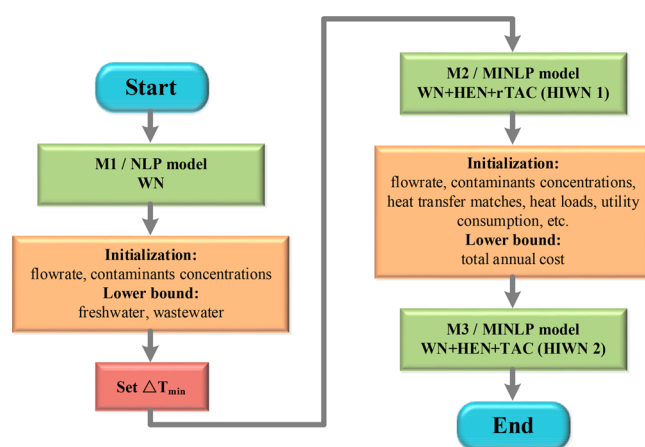


Figure 3. Three-step solution strategy for the synthesis of HIWN.

Matching Composite Curve on H–F Diagram.¹⁷ Besides, conceptual design methods have also been developed to solve multiple contaminants problems. These methods include

Concentration Order and Temperature Composite Curve (COTCC)¹⁸ and Temperature and Concentration Order Composite Curves (TCOCC).^{19,20} Although, conceptual design has the advantage of good graphical visualization and physical insight, it is still difficult to deal with complex systems. To our best knowledge, there is still no conceptual design method for the synthesis of HIWN with wastewater treatment units. Besides, conceptual design methods rely on the designer's experience. However, mathematical programming methods can generate solutions automatically by optimizing single or multiple objective functions, such as total annual cost (TAC).

The sequential mathematical programming method was first introduced by Bagajewicz et al.²¹ Two sequential linear programming (LP) problems were applied to obtain the freshwater consumption and the utility consumption, followed by a mixed-integer linear programming (MILP) transshipment model. However, an artificially merging procedure was needed to get the specific HEN structure. Liao et al.²² proposed a MINLP model to target freshwater consumption, energy consumption for the design of energy efficient water systems allowing operation split. Besides, they²³ developed a two-step systematic approach and an innovative identification method of hot and cold streams in HIWN. In the first step, a transshipment-based MILP model was used to identify the promising matches. Then, the detailed HEN could be obtained by the well-known stage-wise superstructure.²⁴ The proposed model could be applied for both wastewater uniform and separate treating cases. Subsequently, Zhou et al.²⁵ extended the two-step approach to a one-step approach by an improved superstructure, which can be formulated as both the mathematical model with equilibrium constraints (MPEC) and the mixed-integer nonlinear programming (MINLP) model. Recently, Liu et al.²⁶ developed a novel disjunctive model to determine water-using network, where water and energy are simultaneously optimized, followed by pinch technology for the synthesis of HEN. Almaraz et al.²⁷

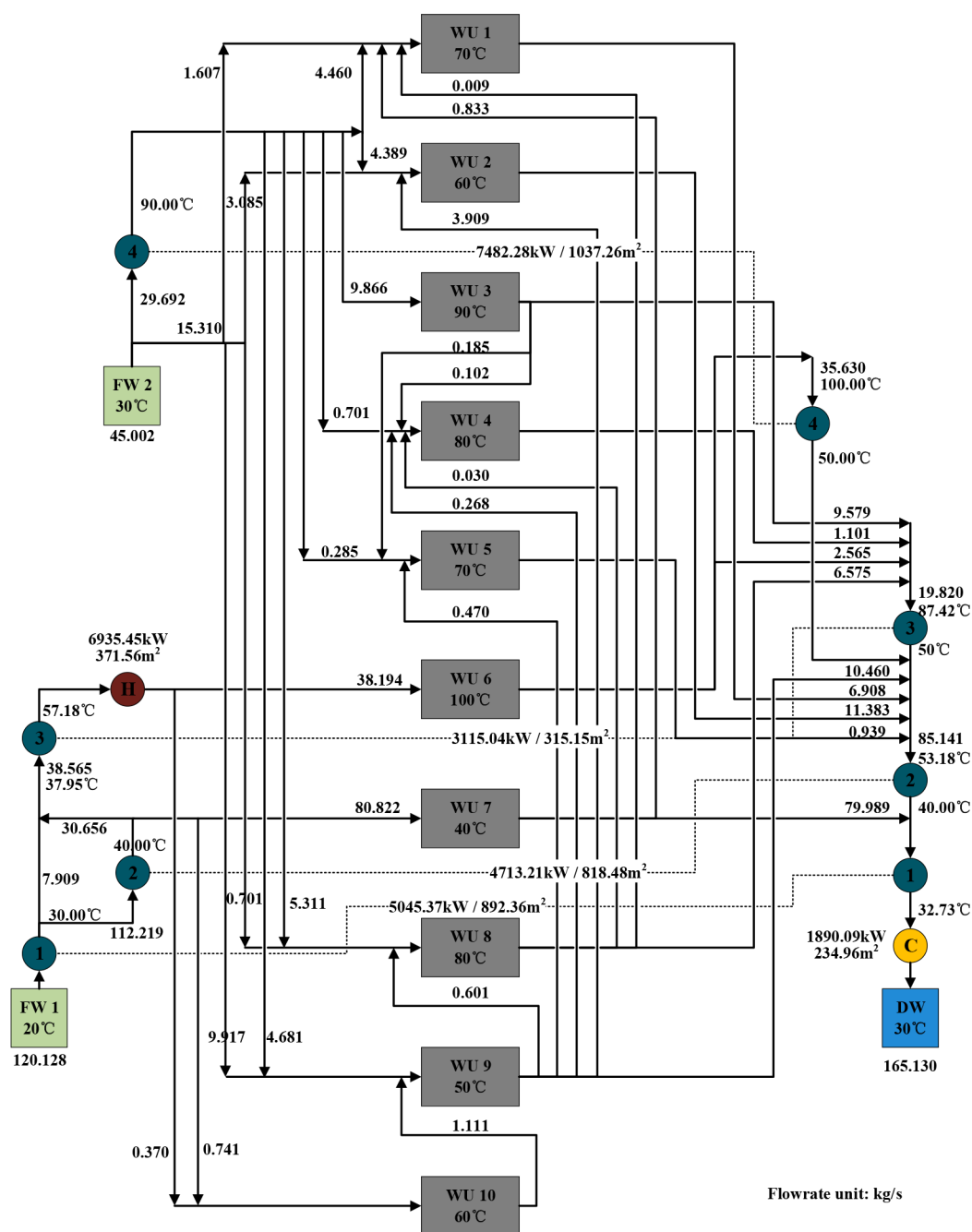


Figure 4. HIWN for example 1.

Table 1. Result Comparisons for Example 1

	Ibrić et al. ⁴⁰	This paper
TAC, \$/y	3138070.0	3123261.8
Freshwater consumption, kg/s	120.128	120.128
Secondary water consumption, kg/s	44.453	45.002
Hot utility consumption, kW	6912.8	6935.45
Cold utility consumption, kW	1867.5	1890.09
Utility cost, \$/y	885561.0	888956.8
Water cost, \$/y	1857867.8	1859445.8
Investment cost for HEs, \$/y	394641.2	374859.2

proposed a two-stage methodology, including the multiobjective optimization of WN and pinch analysis and mathematical programming of HEN. It is important to point out that the

interconnection between subsystems (WN and HEN) cannot be fully explored by sequential methods, which generally lead to suboptimal solutions.

Simultaneous mathematical programming methods can handle the above-mentioned difficulties of sequential methods. Many different superstructure-based models were formulated to address the synthesis of HIWN simultaneously. Bogataj and Bagajewicz²⁸ presented a superstructure combining WN and HEN, based on a modified stage-wise superstructure.²⁴ Dong et al.²⁹ developed a modified state-space superstructure and a hybrid optimization strategy, which could enhance the solution quality and efficiency. Subsequently, a multiscale stage-space superstructure was introduced by Zhou et al.^{30,31} to capture all possible network configurations for the synthesis of integrated interplant water allocation networks and heat exchanger

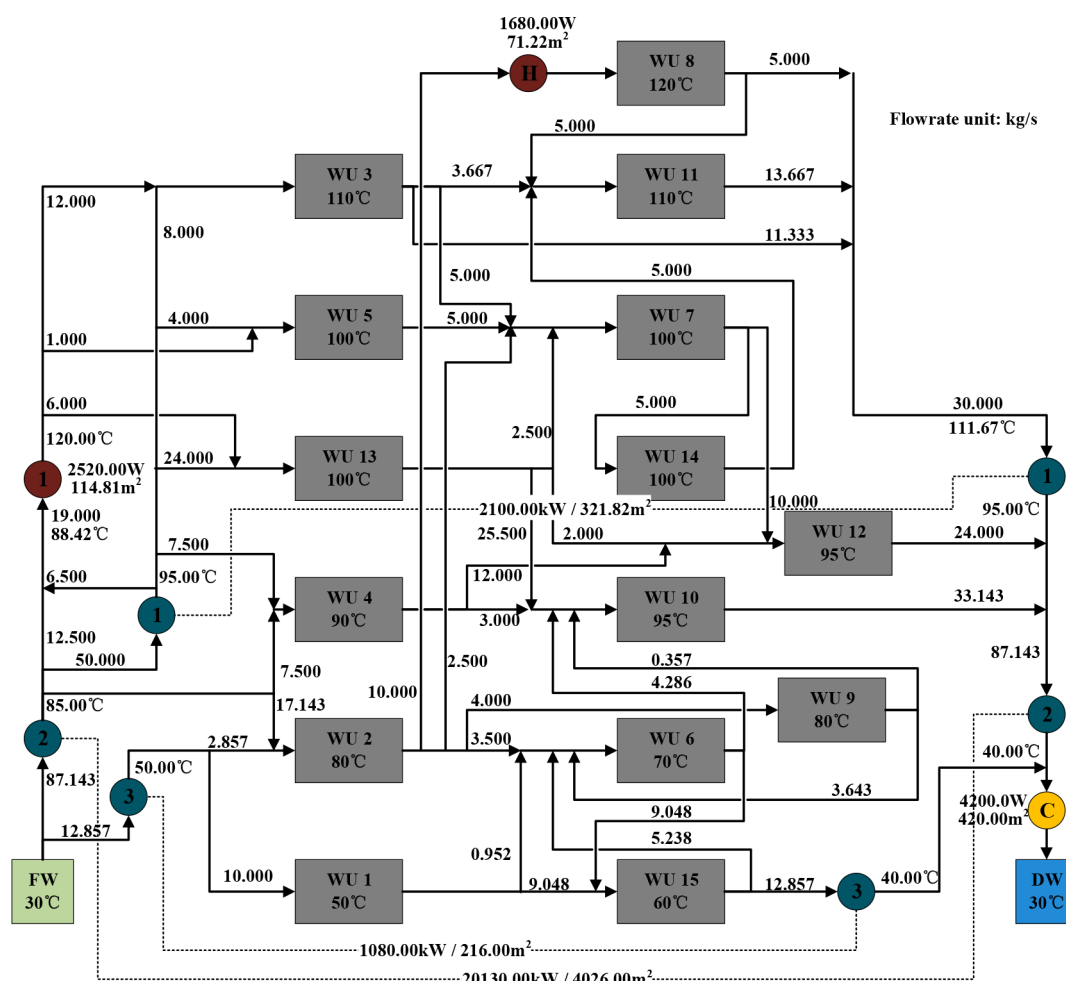


Figure 5. HIWN for example 2.

Table 2. Result Comparisons for Example 2

	Liu et al. ²⁶	Liao et al. ¹⁷	Hong et al. ⁴³	This paper
TAC, \$/y	4027020	3967687	3951376	3876046.5
Freshwater consumption, kg/s	100.000	100.000	100.000	100.000
Hot utility consumption, kW	4200.00	4200.00	4200.00	4200.00
Cold utility consumption, kW	4200.00	4200.00	4200.00	4200.00
Utility cost, \$/y	2423400	2423400	2423400	2423400.0
Investment cost for HEs, \$/y	523619	464287	447976	372646.5

networks (IWAHEN). What is more, Ahmetović and Kravanja³² modified the WN superstructure³³ by incorporating additional opportunities for heat integration. This work was extended by incorporating process-to-process streams in HEN³⁴ and wastewater treatment units.³⁵ Yan et al.³⁶ modified this superstructure by changing the placement of heaters and coolers. The superstructure was formulated as a NLP problem, by using a new methodology to replace binary variables. Ibrić et al. addressed a novel simultaneous model for the synthesis of pinched and threshold HIWN³⁷ and extended the model to include wastewater treatment units.³⁸ Subsequently, a compact superstructure³⁹ was proposed, enabling direct and indirect heat exchange with a manageable number of hot and cold streams. In their subsequent work, a superstructure reduction strategy⁴⁰ with

18 rules was developed to reduce the superstructure's complexity and the overall model size. Hong et al.⁴¹ proposed a novel superstructure featuring parallel HEN structure, which was suitable for both uniform wastewater treatment and separate wastewater cases. Recently, Ghazouani et al.⁴² developed a MILP model to design mass allocation and heat exchanger networks simultaneously. However, it was only applied to the problems with fixed flow rate units.

The mathematical programming is outstanding to solve the overall model for the synthesis of HIWN. However, the model can be very complex, especially for large-scale problems. Among various kinds HIWN problems, the synthesis of the HIWN with fixed mass-load units, multiple contaminants, and wastewater treatment units is the most difficult. There are several published models for the above-mentioned problem. However, to our best knowledge, none of them could be solved to a good solution without an initial point, which were generated by random initial values^{29,31} or running a reduced model.^{35,38,40} It is known that the nonlinearity of the model is one of the reasons why the models are difficult to solve. The nonlinear terms are caused by the nonisothermal mixing in both water networks and heat exchanger networks, as well as the mixing of streams with different contaminant concentrations in water networks. In our previous research,⁴³ the nonlinear terms caused by nonisothermal mixing was omitted by using the modified transshipment model. A MILP model was developed to target the simplified TAC for the synthesis of heat integrated water-using

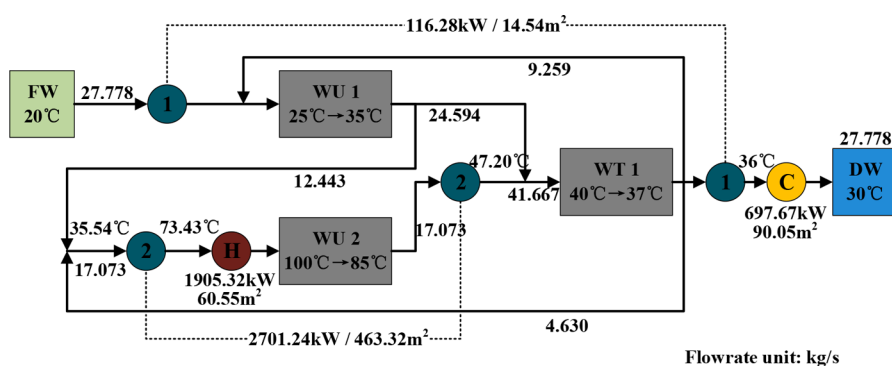


Figure 6. HIWN for example 3.

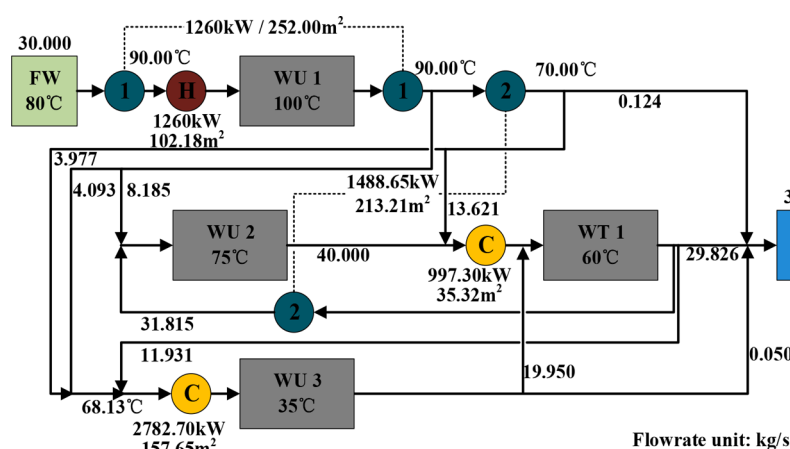
Figure 7. HIWN for example 4 (Case 1, $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$).

Table 3. Result Comparisons for Example 4, Case 1

	Ahmetović et al. ³⁵ $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$	Ibrić et al. ³⁸ $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$	Ibrić et al. ⁴⁰ $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$	Ibrić et al. ⁴⁰ $\Delta T_{\min} = 1\text{ }^{\circ}\text{C}$	This paper $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$	This paper $\Delta T_{\min} = 1.4\text{ }^{\circ}\text{C}$
TAC, \$/y	1821220.3	1816963.8	1798312.0	1314901.0	1812594.1	1317391.1
Freshwater consumption, kg/s	30.000	30.000	30.000	30.000	30.000	30.000
Wastewater treatment, kg/s	73.571	73.571	76.817	76.770	73.571	74.301
Hot utility consumption, kW	1260.00	1260.00	1260.00	177.60	1260.00	176.40
Cold utility consumption, kW	3780.00	3780.00	3780.00	2697.60	3780.00	2696.40
Utility cost, \$/y	1189440.0	1189440.0	1189440.0	576801.6	1189440.0	576122.4
Operating cost for WT, \$/y	14196.3	14196.3	14822.6	14813.5	14196.3	14337.1
Investment cost for WT, \$/y	62586.2	62586.2	64506.4	64478.7	62586.2	63019.9
Investment cost for HEs, \$/y	166197.7	161941.2	140740.7	270007.2	157571.5	275111.6

networks (HIWUN). It is easier to solve in consequence of the linearity of model, however, both multiple contaminants and wastewater treatment units were excluded. In order to make the model more practical for industries, the previous model⁴³ is extended to include multiple contaminants and wastewater treatment units. Owing to the features of the modified transshipment model, the nonlinear terms only exist in the model of the water network. Thus, this characteristic of the proposed model, with less nonlinear terms, would make the model easier to solve. Because the solutions significantly depend on the provided initial point, an effective solution strategy is indispensable for obtaining a good result. In this research, a

three-step solution strategy is developed. The motivation of the first step is to provide initial values for the nonlinear terms in the second step. Then, the rTAC is minimized in the second step, which can consider the trade-offs between the capital cost and the operating cost roughly. In the final step, the TAC is optimized, with the initial point from the second step.

Problem Statement. The general problem of heat integrated water network synthesis can be defined as follows. Given is a set of water-using units (WU) with specified operating temperature, certain contaminant load to be removed, and contaminant concentration constraints for inlet and outlet streams. Besides, a set of wastewater treatment units (WT)

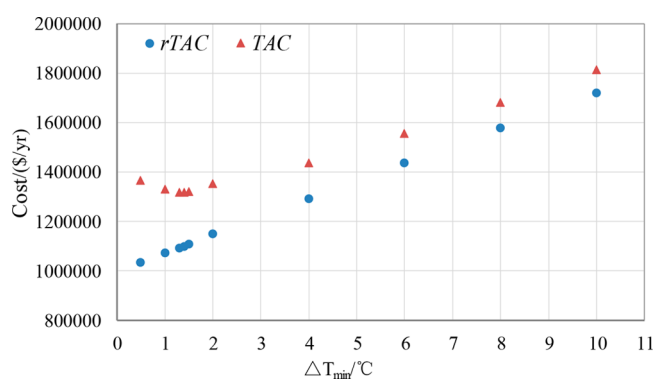


Figure 8. Optional solutions for example 4 (Case 1).

with specified operating temperature is given. The hot and cold utilities (HU, CU) and a set of freshwater sources (FW) with specified temperature and contaminant concentration are given, while the discharge wastewater (DW) stream is specified to be below a maximum temperature. The objective is to determine the freshwater consumption, the utility consumption, the network parameters (include stream flow rates, temperatures, contaminant concentrations), and the network topology (include interconnections among FWs, WUs, and WTs, heat integration between hot streams and cold streams, and the utility distribution) by minimizing the total annual cost.

Several assumptions are given:

- Only one hot utility and one cold utility are available.
- WUs and WTs operate isothermally (Except Example 3) and continuously.
- Heat capacities and heat transfer coefficients of streams are constants.
- Heat exchangers are all counter-current shell and tube heat exchangers.
- Constant removal ratios of contaminants in WTs are given.
- Constant mass loads of contaminants in WUs are given.

Representation of HIWN and Model Formulation. A recently proposed model⁴³ has been modified to include multiple contaminants and wastewater treatment units. In the previous model,⁴³ the identification of hot and cold streams from Liao et al.²³ is adopted. Each WU is considered as two parts: source (S) and demand (D), corresponding to the outlet and inlet of each unit, respectively. Based on this idea, it is assumed that each unit

is connected with one hot source stream l_s (HS, red solid line), one cold source stream m_s (CS, blue solid line), one hot demand stream l_d (HD, red dashed line), and one cold demand stream m_d (CD, blue dashed line). In the proposed model, WTs are included, as shown in Figure 1. It shows the representation of HIWN, including WUN, WTN, and HEN. Similar to the WUs, each WT is also connected with four streams, one hot source stream, one cold source stream, one hot demand stream, and one cold demand stream. Because the freshwater temperature and the discharge temperature are generally lower than the operating temperature of all WUs and WTs, there is only a CS stream for the freshwater source and only a HD stream for the discharge wastewater demand. More details about the identification of hot streams and cold streams can be found in Liao et al.²³ and Hong et al.⁴¹

Based on the identification of hot and cold streams, the modified transshipment model is adopted to represent the HEN. In the HEN, there are mass flows from hot source streams to hot demand streams (purple solid line), as well as mass flows from cold source streams to cold demand streams (green solid line). The mass flows may happen in each temperature levels which are determined by the starting and ending temperatures of hot and cold streams and the exchanger minimum approach temperature ($\text{EMAT}/\Delta T_{\min}$). It is worth mentioning that the mass flow from one source stream to one demand stream actually represents the mass flow from the outlet of one unit to the inlet of another unit. Thus, the mass flows between different units in the WUN and WTN, including the reuse of water, the regeneration reuse of water, the freshwater supply, and the wastewater discharge, are all embedded in the HEN by the mass flows from source streams to demand streams. One of the significant differences between the proposed model and most of reported superstructures^{28,32,34,35,37,38,41} is that there is no need to construct specific superstructures for WUN and WTN. Introducing the wastewater units into the previous model⁴³ does not change the mass flow pattern, where the hot streams from WUs and WTs are treated equally. However, introducing WTs will change the model formulations, owing to the different principles between these two kinds of units. The model formulations will be introduced in the next section.

Water-Using and Wastewater Treatment Network. Compared to our recently proposed model,⁴³ multiple contaminants and wastewater treatment units are considered in this model, which will lead to nonlinear formulations. The main difference

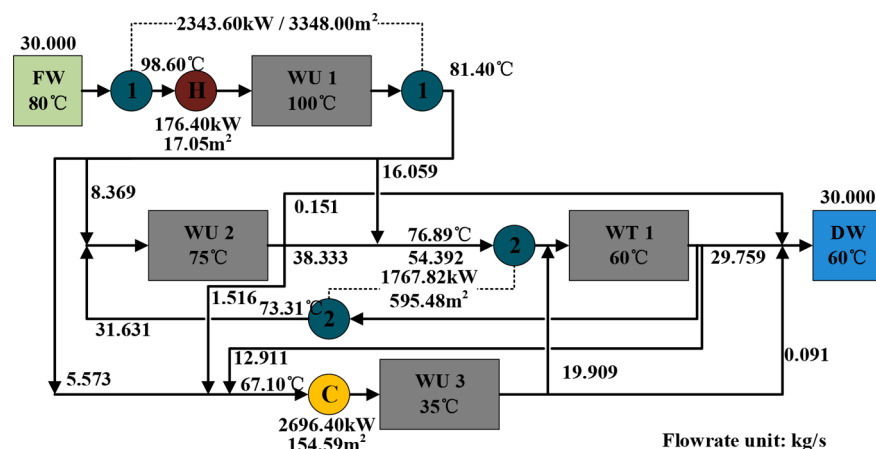


Figure 9. HIWN for example 4 (Case 1, $\Delta T_{\min} = 1.4^{\circ}\text{C}$).

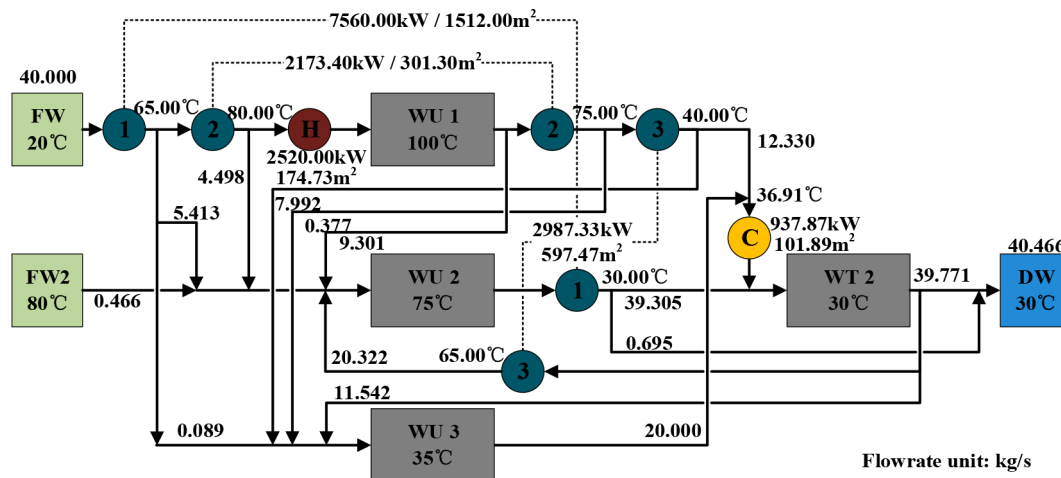


Figure 10. HIWN for example 4 (Case 2, $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$).

Table 4. Result Comparisons for Example 4, Case 2

	Ibrić et al. ³⁸	This paper
TAC, \$/y	1925101.1	1915940.6
Freshwater consumption (20 °C), kg/s	28.094	40.000
Freshwater consumption (80 °C), kg/s	1.906	0.466
Wastewater treatment 2, kg/s	73.736	71.635
Wastewater treatment 3, kg/s	30.501	0
Hot utility consumption, kW	2420.96	2520.00
Cold utility consumption, kW	1641.31	937.87
Freshwater cost, \$/y	328117.0	438040.0
Utility cost, \$/y	1222909.5	1127297.6
Operating cost for WT, \$/y	14228.1	13822.7
Investment cost for WT, \$/y	75557.0	61428.5
Investment cost for HEs, \$/y	284289.5	275351.7

between WUs and WTs lies on the formulation to calculate the contaminant concentrations of the outlet streams, which are given by eqs 1 and 2, respectively.

Contaminant concentration of outlet stream of each WU:

$$\sum_{i' \in S} sd_{i',j} \times ciout_{i,c} = \sum_{i' \in S} (sd_{i',j} \times ciout_{i',c}) + load_{i,c} \times 1000 \quad \forall i \in \text{WU}, c \in C, i = j \quad (1)$$

Contaminant concentration of outlet stream of each WT:

$$\sum_{i' \in S} sd_{i',j} \times ciout_{i,c} = \sum_{i' \in S} (sd_{i',j} \times ciout_{i',c}) \times (1 - rr_{i,c}) \quad \forall i \in \text{WT}, c \in C, i = j \quad (2)$$

Maximum inlet concentration constraint for WUs and discharged wastewater:

$$\sum_{i \in S} (sd_{i,j} \times ciout_{i,c}) \leq f_j \times cjinm_{j,c} \quad \forall j \in \text{WU} \cap \text{DW}, c \in C \quad (3)$$

The other formulations of the WUN and WTN are as follows. Besides, lower and upper bounds of variables are given in the Supporting Information (eqs S1–S12), to speed up the solving process.

Inlet and outlet mass balance of each unit:

$$f_i = f_j \quad \forall i \in S, j \in D, i \notin \text{FW}, j \notin \text{DW}, i = j \quad (4)$$

Mass balance of each source:

$$f_i = \sum_{j \in D} sd_{i,j} \quad \forall i \in S \quad (5)$$

Mass balance of each demand:

$$f_j = \sum_{i \in S} sd_{i,j} \quad \forall j \in D \quad (6)$$

Freshwater consumption:

$$fresh = \sum_{i \in \text{FW}} f_i \quad (7)$$

Discharge wastewater:

$$waste = \sum_{j \in \text{DW}} f_j \quad (8)$$

Mass balance between freshwater and wastewater:

$$fresh = waste \quad (9)$$

Mass Flow Pattern and Heat Flow Pattern. The first step to describe the mass flow pattern is to construct the temperature levels n ($n \in \text{TL}$) according to the starting and ending temperatures of hot and cold streams. Once the temperature levels are constructed, the location of each hot and cold stream among the whole temperature intervals k ($k \in \text{TL}$) can be obtained, because the temperature bounds of each stream are known. Mass flows only happen on the determined temperature levels. Because each hot (cold) stream may exchange heat with more than one cold (hot) streams in one temperature interval, each hot (cold) stream l (m) should be split into several substreams ls (ms) in the mass flow pattern. The formulations of the mass flow pattern can be found in the Supporting Information (eqs S13–S42).

With the constructed temperature intervals and the location of each hot and cold stream, the modified^{43,44} transshipment model of Papoulias and Grossmann⁴⁵ is adopted to describe the heat flow pattern. Isothermal mixing and nonisothermal mixing can happen in the temperature levels by mass flows from source streams to demand streams and mass flows between substreams, whereas the indirect heat transfers can happen in each temperature interval. The formulations of heat flow pattern are as follows. Lower and upper bounds of variables are given in the Supporting Information (eqs S43–S54).

Energy balance at each temperature interval:

$$rqhs_{l,ls,n+1} - rqhs_{l,ls,n} + \sum_{(m,ms) \in CMS, m \in C_k} q_{l,ls,m,ms,k} = fls$$

$$k_{l,ls,k} \times (tn_n - tn_{n+1}) \times c_p$$

$$\forall (l, ls) \in HLS, n \in TL, k \in TI, n = k, ns_l \leq k < ne_l \quad (10)$$

$$rqhs_{l,ls,n} = \begin{cases} 0 & \forall (l, ls) \in HLS, n \in TL, n \leq ns_l \text{ or } n > ne_l \\ 0 & \forall (l, ls) \notin HLS, n \in TL \end{cases} \quad (11)$$

$$rqcs_{m,ms,n} - rqcs_{m,ms,n+1} + \sum_{(l,ls) \in HLS, l \in H_k} q_{l,ls,m,ms,k} = fms$$

$$k_{m,ms,k} \times (tn_n - tn_{n+1}) \times c_p$$

$$\forall (m, ms) \in CMS, n \in TL, k \in TI, n = k, ne_m < n < ns_m \quad (12)$$

$$rqcs_{m,ms,n} = \begin{cases} 0 & \forall (m, ms) \in CMS, n \in TL, n < ne_m \text{ or } n \geq ns_m \\ 0 & \forall (m, ms) \notin CMS, n \in TL \end{cases} \quad (13)$$

Energy balance for demand streams:

$$hhd_d = \sum_{ls \in LS} rqhs_{l_d,ls,ne_{l_d}} - qc_{l_d} - \sum_{l_s \in HS} \sum_{n > ne_{l_d}} trh_{l_s,l_d,n} \times c_p$$

$$\times (the_{l_d} - tn_n) \quad \forall l_d \in HD \quad (14)$$

$$hcd_{m_d} = \sum_{ms \in MS} rqcs_{m_d,ms,ne_{m_d}} - qh_{m_d}$$

$$- \sum_{m_s \in CS} \sum_{n < ne_{l_d}} trh_{m_s,m_d,n} \times c_p \times (tn_n - tce_{m_d}) \quad \forall m_d \in CD \quad (15)$$

$$hhd_d - hcd_{m_d} = 0 \quad \forall l_d \in HD, m_d \in CD, l_d = m_d \quad (16)$$

Energy balance for source streams:

$$\sum_{ls \in LS} rqhs_{l_s,ls,ne_{l_s}} = qc_{l_s} \quad \forall l_s \in HS \quad (17)$$

$$\sum_{ms \in MS} rqcs_{m_s,ms,ne_{m_s}} = qh_{m_s} \quad \forall m_s \in CS \quad (18)$$

Logical constraints:

$$q_{l,ls,m,ms,k} - \Omega_{l,m,k} \times z_{l,ls,m,ms,k} \leq 0$$

$$\forall (l, ls) \in HLS, (m, ms) \in CMS, k \in TI, ns_l \leq k < ne_l, n$$

$$e_m \leq k < ns_m \quad (19)$$

$$qh_m - CL_m \times zh_m \leq 0 \quad \forall m \in C \quad (20)$$

$$qc_l - HL_l \times zc_l \leq 0 \quad \forall l \in H \quad (21)$$

$$rqhs_{l,ls,n} \leq MRLn_{l,n} \times (1 - mlsk_{l,ls,k})$$

$$\forall (l, ls) \in HLS, n \in TL, k \in TI, n = k, ns_l < k < ne_l \quad (22)$$

$$rqcs_{m,ms,n} \leq MRMn_{m,n} \times (1 - mmsk_{m,ms,k})$$

$$\forall (m, ms) \in CMS, n \in TL, k \in TI, n = k, ne_m < k < ns_m \quad (23)$$

$$zc_l + zh_m \leq 1 \quad \forall l \in HD, m \in CD, l = m \quad (24)$$

$$\sum_{(m,ms) \in CMS} z_{l,ls,m,ms,k} \leq 1 \quad \forall (l, ls) \in HLS, k \in TI, ns_l \leq k < ne_l \quad (25)$$

$$\sum_{(l,ls) \in HLS} z_{l,ls,m,ms,k} \leq 1 \quad \forall (m, ms) \in CMS, k \in TI, ne_m \leq k < ns_m \quad (26)$$

Compared to the widely adopted HEN superstructure, namely stage-wise superstructure,²⁴ the most important advantage of the transshipment model is that the model can keep constraints linear while allowing nonisothermal mixing. Besides, several alternative structures neglected by the stage-wise superstructure²⁴ are considered in the modified transshipment model.^{43,44} An illustration for nonisothermal mixing is shown in Figure 2-1, where a stream of 40 °C is merged with another stream in temperature level 40 °C. Note that the temperature of the other stream is a variable that is above 40 °C, while the flow rate of both streams are also variables. Consequently, the product of the temperature and flow rate of the other stream is a nonlinear term. Such a nonlinear term can be replaced by the residual energy concept ($rqhs_{l,ls,n}$) of the transshipment model, which represents the available energy in the temperature level n . Therefore, we obtain linear energy balance equations as shown by eqs 10–13. Furthermore, nonisothermal mixing also exists at the inlet of each unit in the water network. It is known that there is one demand hot stream and one demand cold stream at the inlet, as shown in Figure 2-2. Generally, one nonlinear term is needed to make sure that the temperature of the mixing stream is equal to the operating temperature of the unit. However, the nonlinear term can be omitted owing to the residual energy concept. As shown in eqs 14–16, the residual energy of the hot demand stream must be equal to the one of the cold demand stream.

Identification of Heat Exchangers and Temperature Difference for Heat Transfer. The traditional transshipment model⁴⁵ only indicates the heat flow pattern, including the heat transfer matches and the corresponding heat loads. Nonetheless, the heat exchanger network structure cannot be straightly constructed without the mass flow pattern in the HIWN. In the modified transshipment model,⁴³ the heat flow pattern and the mass flow pattern are both included. A set of formulations is developed to identify all heat exchangers in the heat flow pattern. The readers can refer to our recent research⁴³ and the formulations in the Supporting Information (eqs S55–S62).

In our previous model,⁴³ EMAT is adopted as the temperature difference for each heat transfer to keep the model linear. However, in that case, the heat exchanger area cannot be accurately calculated and then optimized. Thus, in order to fully consider the trade-offs between the capital cost and the operating cost, the accurate temperature difference is considered in this model. The inlet and outlet temperatures of hot (cold) substream l/ls (m/ms) in temperature interval k are given by eqs 27–30, while the temperature differences for all heat transfers are given by eqs 31–33. It should be noted that eqs 27–33 are all nonlinear and nonconvex. In order to reduce the nonlinearity of constraints, the temperature difference for each heat transfer is substituted into the objective function.

$$tih_{l,ls,k} = tn_n + \Delta T_{\min} + \frac{rqhs_{l,ls,n}}{flsk_{l,ls,k} \times c_p}$$

$$\forall (l, ls) \in HLS, k \in TI, ns_l \leq k < ne_l \quad (27)$$

$$toh_{l,ls,k} = tn_{n+1} + \Delta T_{\min} + \frac{rqhs_{l,ls,n+1}}{flsk_{l,ls,k} \times c_p}$$

$$\forall (l, ls) \in HLS, k \in TI, ns_l \leq k < ne_l \quad (28)$$

$$tic_{m,ms,k} = tn_{n+1} - \frac{rqhs_{m,ms,n+1}}{fmsk_{m,ms,k} \times c_p}$$

$$\forall (m, ms) \in CMS, k \in TI, ne_m \leq k < ns_m \quad (29)$$

$$toc_{m,ms,k} = tn_n - \frac{rqhs_{m,ms,n}}{fmsk_{m,ms,k} \times c_p}$$

$$\forall (m, ms) \in CMS, k \in TI, ne_m \leq k < ns_m \quad (30)$$

$$dt_{l,ls,m,ms,k} = [(tih_{l,ls,k} - toc_{m,ms,k})(toh_{l,ls,k} - tic_{m,ms,k}) \times 0.5 \\ \times (tih_{l,ls,k} - toc_{m,ms,k} + toh_{l,ls,k} - tic_{m,ms,k})]^{1/3}$$

$$= \left[\left(\Delta T_{\min} + \frac{rqhs_{l,ls,n}}{flsk_{l,ls,k} \times c_p} + \frac{rqhs_{m,ms,n}}{fmsk_{m,ms,k} \times c_p} \right) \right. \\ \times \left(\Delta T_{\min} + \frac{rqhs_{l,ls,n+1}}{flsk_{l,ls,k} \times c_p} + \frac{rqhs_{m,ms,n+1}}{fmsk_{m,ms,k} \times c_p} \right) \times 0.5 \\ \times \left(\Delta T_{\min} + \frac{rqhs_{l,ls,n}}{flsk_{l,ls,k} \times c_p} + \frac{rqhs_{m,ms,n}}{fmsk_{m,ms,k} \times c_p} + \Delta T_{\min} \right. \\ \left. \left. + \frac{rqhs_{l,ls,n+1}}{flsk_{l,ls,k} \times c_p} + \frac{rqhs_{m,ms,n+1}}{fmsk_{m,ms,k} \times c_p} \right) \right]^{1/3}$$

$$\forall (l, ls) \in HLS, (m, ms) \in CMS, k \in TI, ns_l \leq k < ne_l, n \\ e_m \leq k < ns_m \quad (31)$$

$$dteu_l = \left[\left(the_l + \Delta T_{\min} + \frac{\sum_{ls} rqhs_{l,ls,ne_l}}{flk_{l,k(k=ne_l-1)} \times c_p} - \frac{qc_l}{flk_{l,k(k=ne_l-1)} \times c_p} \right. \right. \\ \left. \left. - ticu \right) \times \left(the_l + \Delta T_{\min} + \frac{\sum_{ls} rqhs_{l,ls,ne_l}}{flk_{l,k(k=ne_l-1)} \times c_p} - tocu \right) \right. \\ \left. \times 0.5 \times \left(the_l + \Delta T_{\min} + \frac{\sum_{ls} rqhs_{l,ls,ne_l}}{flk_{l,k(k=ne_l-1)} \times c_p} \right. \right. \\ \left. \left. - \frac{qc_l}{flk_{l,k(k=ne_l-1)} \times c_p} - ticu + the_l + \Delta T_{\min} \right. \right. \\ \left. \left. + \frac{\sum_{ls} rqhs_{l,ls,ne_l}}{flk_{l,k(k=ne_l-1)} \times c_p} - tocu \right) \right]^{1/3}$$

$$\forall l \in H \quad (32)$$

$$dthu_m = \left[\left(tihu - tce_m + \frac{\sum_{ms} rqs_{m,ms,ne_m}}{fmsk_{m,k(k=ne_m)} \times c_p} - \frac{qh_m}{fmsk_{m,k(k=ne_m)} \times c_p} \right) \right. \\ \times \left(tohu - tce_m + \frac{\sum_{ms} rqs_{m,ms,ne_m}}{fmsk_{m,k(k=ne_m)} \times c_p} \right) \times 0.5 \\ \times \left(tihu - tce_m + \frac{\sum_{ms} rqs_{m,ms,ne_m}}{fmsk_{m,k(k=ne_m)} \times c_p} \right. \\ \left. - \frac{qh_m}{fmsk_{m,k(k=ne_m)} \times c_p} + tohu - tce_m \right. \\ \left. \left. + \frac{\sum_{ms} rqs_{m,ms,ne_m}}{fmsk_{m,k(k=ne_m)} \times c_p} \right) \right]^{1/3}$$

$$\forall m \in C \quad (33)$$

Objective Functions. Owing to the nonlinearity and non-convexity of the proposed model, it is significantly difficult to get a good solution without a good initial point. It is known that the formulations (eqs 1 and 2) to calculate the contaminant concentrations of outlet streams of each unit in the WN are nonlinear and nonconvex. Except that, the nonlinear items only exist in the objective function (total annual cost, eq 35), caused by the formulations for the capital cost of heat exchangers and wastewater treatment units. In consideration of the characteristics of this model, a three-step solution strategy (see Figure 3) is

developed to solve the model, ensuring obtaining promising solutions. Three models, named **M1**, **M2**, and **M3**, are developed, which are written as

$$\min \text{fresh}$$

$$\text{s.t. eqs 1–9, S1–S2} \quad (M1)$$

$$\min rTAC$$

$$\text{s.t. eqs 1–9, 10–26, 36–38, S1–S62} \quad (M2)$$

$$\min TAC$$

$$\text{s.t. eqs 1–9, 10–26, 27–33, S1–S62} \quad (M3)$$

Where the objective functions of **M2** and **M3** are defined as follows:

$$rTAC = \sum_{i \in FW} (CFW_i \times H \times f_i) + \sum_m qh_m \times CHU \\ + \sum_l qc_l \times CCU + \sum_{i \in WTS} (CWT_i \times H \times f_i) \\ + AF \times rCT_i \times \sum_{i \in WTS} f_i + CF_{l,m} \times \\ \sum_{l,ls \in HLS} \sum_{m,ms \in CMS} \sum_{k \in TI} zz_{l,ls,m,ms,k} + CF_{l,CU} \times \sum_{l \in H} zc_l \\ + CF_{m,HU} \times \sum_{m \in C} zh_m + rC_{l,m} \times \\ \sum_{(l,ls) \in HLS} \sum_{(m,ms) \in CMS} \sum_{ns_l \leq k < ns_m} \frac{q_{l,ls,m,ms,k}}{U_{l,m} \times rdt_{l,ls,m,ms,k}} \\ + rC_{l,CU} \times \sum_{l \in H} \frac{qc_l}{U_{l,CU} \times rdtcu_l} + rC_{m,HU} \times \\ \sum_{m \in C} \frac{qh_m}{U_{m,HU} \times rdthum} \quad (34)$$

$$TAC = \sum_{i \in FW} (CFW_i \times H \times f_i) + \sum_m qh_m \times CHU + \sum_l qc_l \times CCU \\ + \sum_{i \in WTS} (CWT_i \times H \times f_i) + AF \times CT_i \times \sum_{i \in WTS} (f_i)^{\alpha_i} \\ + CF_{l,m} \times \sum_{l,ls \in HLS} \sum_{m,ms \in CMS} \sum_{k \in TI} zz_{l,ls,m,ms,k} \\ + CF_{l,CU} \times \sum_{l \in H} zc_l + CF_{m,HU} \times \sum_{m \in C} zh_m \\ + C_{l,m} \times \sum_{(l,ls) \in HLS} \sum_{(m,ms) \in CMS} \left(\sum_{ns_l \leq k < ns_m} \frac{q_{l,ls,m,ms,k}}{U_{l,m} \times dt_{l,ls,m,ms,k}} \right)^{B_{l,m}} \\ + C_{l,CU} \times \sum_{l \in H} \left(\frac{qc_l}{U_{l,CU} \times dtcu_l} \right)^{B_{l,CU}} + C_{m,HU} \times \\ \sum_{m \in C} \left(\frac{qh_m}{U_{m,HU} \times dthum} \right)^{B_{m,HU}} \quad (35)$$

The total annual cost (TAC, eq 35) includes both the operating cost and the capital cost. The terms in the first row of eq 35 represent the freshwater cost, the hot utility cost, and the cold utility cost, respectively. The terms in the second row represent the operating cost and the capital cost of wastewater treatment units. The remaining terms represent the capital cost of heat exchangers, heaters, and coolers. As mentioned before, the formulations of temperature differences are nonlinear, which are all substituted into eq 35. In order to circumvent this problem, relaxed temperature differences are proposed in this model, which are formulated as linear terms, given by eqs 36–38. Thus, a relaxed objective can be generated and optimized before optimizing the accurate objective. In the relaxed total annual cost

(rTAC, eq 34), the temperature differences for heat transfers are replaced by the relaxed temperature differences (eqs 36–38). The inlet and outlet temperatures of hot (cold) streams in each temperature level are both replaced by the corresponding upper (lower) bounds, which are the starting temperature of hot (cold) streams. Consequently, the temperature difference of each heat transfer is overestimated, which makes the heat exchanger area underestimated. The proof procedures are given, following eqs 36–38, respectively. Besides, the exponential cost formulations of wastewater treatment units and heat exchangers are linearized, according to the upper bound of wastewater and the upper bound of each heat transfer area.

$$\begin{aligned} rdt_{l,ls,m,ms,k} &= [(ths_l - tcs_m)(ths_l - tcs_m) \times 0.5 \times (ths_l - tcs_m + ths_l - tcs_m)]^{1/3} \\ &= (ths_l - tcs_m) \quad \forall (l, ls) \in \text{HLS}, (m, ms) \in \text{CMS}, k \in \text{TI}, n \\ s_l \leq k < n_{e_l}, n_{e_m} \leq k < n_{s_m} \end{aligned} \quad (36)$$

$$\therefore ths_l \geq tih_{l,ls,k} \& ths_l \geq toh_{l,ls,k} \& tcs_m \leq tic_{m,ms,k} \& tcs_m \leq toc_{m,ms,k}$$

$$\therefore rdt_{l,ls,m,ms,k} \geq dt_{l,ls,m,ms,k}$$

$$\begin{aligned} rdtcu_l &= [(ths_l + \Delta T_{\min} - ticu) \times (ths_l + \Delta T_{\min} - tocu) \times 0.5 \\ &\quad \times (ths_l + \Delta T_{\min} - ticu + ths_l + \Delta T_{\min} - tocu)]^{1/3} \\ \forall l \in \text{H} \end{aligned} \quad (37)$$

$$\begin{aligned} \therefore ths_l &\geq the_l + \frac{\sum_{ls} rqhs_{l,ls,n_{e_l}}}{fkk_{l,k(k=n_{e_l}-1)} \times c_p} \\ &\geq the_l + \frac{\sum_{ls} rqhs_{l,ls,n_{e_l}}}{fkk_{l,k(k=n_{e_l}-1)} \times c_p} - \frac{qc_l}{fkk_{l,k(k=n_{e_l}-1)} \times c_p} \end{aligned}$$

$$\therefore rdtcu_l \geq dtcu_l$$

$$\begin{aligned} rdthu_m &= [(tihu - tcs_m) \times (tohu - tcs_m)] \times 0.5 \quad \forall m \in \text{C} \\ &\quad \times (tihu - tcs_m + tohu - tcs_m)]^{1/3} \end{aligned} \quad (38)$$

$$\begin{aligned} \therefore tcs_m &\leq tce_m + \frac{\sum_{ms} rqc_{m,ms,n_{e_m}}}{fkm_{m,k(k=n_{e_m})} \times c_p} - \frac{qh_m}{fkm_{m,k(k=n_{e_m})} \times c_p} \\ &\leq tce_m + \frac{\sum_{ms} rqc_{m,ms,n_{e_m}}}{fkm_{m,k(k=n_{e_m})} \times c_p} \end{aligned}$$

$$\therefore rdthu_m \geq dthu_m$$

Solution Strategy. In the first step of the three-step solution strategy, the water-using and wastewater treatment network model (M1) is solved by minimizing the freshwater consumption. Then, the heat integrated water network model (M2) with the rTAC is solved by minimizing rTAC. Because the relaxed objective function is linear, the nonlinear terms of model M2 only exist in the water-using and wastewater treatment network model, which reduces the degree of nonlinearity of model M2 and makes it much easier to solve. Besides, the trade-offs between the operating cost and the underestimated capital cost can be considered, resulting in a good initial point for the next step. The obtained rTAC can be the lower bound for the TAC. Note that, when model M2 in the second step can be directly solved, the first step can be omitted. The motivation of the first step is providing initial values for the nonlinear terms in the second step. In the last step, the heat integrated water network model (M3) with accurate TAC is optimized with the initial point from the second step.

Case Study. In this section, four examples with different complexities are given to illustrate the applicability of the proposed model. Two of them are medium HIWN problems with WTs, whereas the other two are large-scale problems with ten and 15 WUs respectively, but without WTs. The data of WUs and WTs and the operating and cost parameters for each example are given in Tables S1–S5 of the [Supporting Information](#).

The proposed model is modeled in GAMS 23.3 (General Algebraic Modeling System) and solved on a PC machine (examples 3 and 4) with 3.2 GHz Intel CPU and 4 GB of RAM and a server (examples 1 and 2) with 2.6 GHz Intel Xeon CPU*2 and 32 GB of ROM. In the first step, COINOPT is adopted as NLP solver, whereas DICOPT with CONOPT and CPLEX is used as MINLP solver in the next two steps.

Example 1. In the first example, a four-contaminants problem⁴⁰ with ten WUs is presented, which includes two water sources: one freshwater source at 20 °C and one secondary water source at 30 °C with contaminants concentration 10 ppm. In order to compare the TAC between the obtained result and the literature result,⁴⁰ the cost data from Ibric et al.⁴⁰ is adopted. It is assumed that only cold streams from two water sources and hot streams to wastewater are involved in the HEN. Besides, the first step to minimize the freshwater consumption is omitted in this example. The optimized result is shown in Figure 4, with the TAC of 3123261.8 \$/y, the freshwater consumption of 120.128 kg/s, the secondary water consumption of 45.002 kg/s, the hot utility consumption of 6935.45 kW, and the cold utility consumption of 1890.09 kW. Compared to Ibric et al.⁴⁰ (Table 1), the freshwater consumption keeps the same. Both Figure 4 and the result in Ibric et al.⁴⁰ indicate that the freshwater is only consumed by WUs 6, 7 and 10, where the maximum inlet concentration for one of contaminants is zero. Besides, more secondary water (45.002 vs 44.453 kg/s) is consumed by other WUs in the obtained result. Although the operating cost increases owing to consuming more secondary water and utility, the capital cost and the TAC decreased 0.50% and 0.47% respectively.

Example 2. The second example presents a large-scale single contaminant problem with 15 WUs from Liu et al.,²⁶ which is the largest problem among all literature HIWN examples. The temperature of freshwater and the discharged wastewater are both 30 °C. This example was also studied in Liu et al.²⁶ and our recent research,^{17,43} where the $\Delta T_{\min}$ of 10 °C was adopted. By the proposed model, the optimized result with TAC of 3876046.5 \$/y is obtained for $\Delta T_{\min} = 10$ °C, as shown in Figure 5. Table 2 provides the comparisons between literature results and Figure 5. The obtained TAC decreases 3.7%, 2.3%, and 1.9% compared to the ones in Liu et al.,²⁶ Liao et al.,¹⁷ and Hong et al.⁴³ respectively. Sequential approaches were adopted in both Liu et al.²⁶ and Liao et al.,¹⁷ where the T-H diagram and the H-F diagram were used to design HEN respectively. The trade-off between the operating cost and the capital cost cannot be fully explored by sequential approaches. Although the HEN and the WUN were simultaneously synthesized in Hong et al.,⁴³ a simplified TAC was optimized rather than the accurate TAC. The research only focus on the targeting of TAC. In this research, the accurate TAC is minimized, with the initial point generated by optimizing the rTAC.

Example 3. In the previous two sections, two large-scale problems are illustrated to demonstrate the capabilities of the proposed model to solve large-scale problems. In the following, two medium examples with WTs are given. The third example^{35,38} presents a three-contaminants problem with two

Table 5. Results and Computational Time for Examples 1–4

		Example 1	Example 2	Example 3	Example 4, Case 1	Example 4, Case 2
Step 1	FW (kg/s)	/	100	15.427	30	40
	Time (s)	/	0.354	0.309	0.300	0.349
Step 2	rTAC (\$/yr)	2845362.3	3607524.5	4735135.0	1720531.1	1735822.9
	cTAC (\$/yr)	3147049.7	3878752.7	5038792.2	1815586.0	1917016.8
	lErrorl (%)	9.6	7.0	6.0	5.2	9.5
	Time (s)	7227.749	12938.306	1.240	16.785	20.974
Step 3	TAC (\$/yr)	3123261.8	3876046.5	5038792.2	1812594.1	1915940.6
	Time (s)	19.162	214.459	1.021	5.483	2.301

Table 6. Results and Computational Time for Examples 1–4 when Minimizing Operating Cost in the Second Step

		Example 1	Example 2	Example 3	Example 4, Case 1	Example 4, Case 2
Step 1	FW (kg/s)	/	100	15.427	30	40
	Time (s)	/	0.264	0.986	0.294	0.330
Step 2	OC (\$/yr)	2739073.5	3503400.0	3866418.8	1592436.3	1577404.9
	cTAC (\$/yr)	3578791.2	4666098.8	5126279.4	1976129.3	2234177.4
	Time (s)	10.110	274.710	3.412	2.255	5.286
Step 3	TAC (\$/yr)	3122868.993	3883874.428	5038792.2	1821853.5	1938421.6
	Time (s)	18.336	291.897	1.092	12.435	11.920

WUs and one WT. The maximum concentration of the discharged wastewater is set to 20 ppm for all three contaminants A, B, and C. The optimized result is obtained for $\Delta T_{\min} = 10^\circ\text{C}$, as shown in Figure 6. In the first step, the minimum freshwater consumption of 15.427 kg/s is obtained. Then, the lower bound for TAC of 4735135.0 \$/y is obtained when minimizing the rTAC. In the final step, the TAC is optimized with the initial point from the second step. The TAC of the obtained network is 5038792.2 \$/y, with the freshwater consumption of 27.778 kg/s, the hot utility consumption of 1905.32 kW, the cold utility consumption of 697.67 kW, and the wastewater treatment of 41.667 kg/s. This example is also optimized by Ahmetović et al.³⁵ and Ibrić et al.³⁸ It should be noted that the obtained result is exactly the same as the results in the reference.^{35,38} The network structure and all parameters of the HIWAN are all the same.

Example 4. The last example is presented by Ahmetović et al.,³⁵ which is an extension (case 1) of the three-contaminants HIWN problem from Dong et al.²⁹ by involving a WT. Note that the temperature of freshwater and discharge wastewater are 80 and 60 °C respectively. Besides, the maximum concentration of three contaminants are all set to 30 ppm. In order to compare the obtained result with the reference results,^{35,38,40} the $\Delta T_{\min}$ of 10 °C (Case 1) is assumed. The HIWAN with TAC of 1812594.1 \$/y is obtained, as shown in Figure 7. From the result comparisons (Table 3), it can be found that the obtained result is the second best with the TAC decrease of 0.47% and 0.24%, compared to the results in Ahmetović et al.³⁵ and Ibrić et al.³⁸ respectively. However, Ibrić et al.⁴⁰ obtained a better result with TAC of 1798312.0 \$/y, using a reduced superstructure based on their previous research.³⁸ Besides, this example is solved repeatedly with different $\Delta T_{\min}$: 8 °C, 6 °C, 4 °C, 2 °C, 1.5 °C, 1.4 °C, 1.3 °C, 1.0 °C, 0.5 °C, to fully explore the trade-offs between the investment cost and the operating cost. The results are shown in Figure 8, including the rTAC from the second step and the TAC from the third step. It can be seen the best result (1317391.1 \$/y) is obtained when $\Delta T_{\min}$ is set to 1.4 °C (Case 2), as shown in Figure 9. In Ibrić et al.,⁴⁰ a better result with a smaller TAC of 1314901.0 \$/y was obtained, when the $\Delta T_{\min}$ was set to 1 °C. However, the minimum temperature difference in their result is 1.41 °C, which is close to the one in our result.

The main reason why the results in Ibrić et al.⁴⁰ cannot be obtained by the proposed model is that the temperature intervals are fixed in the proposed model. The stream splitting only happens in the constructed temperature levels. In spite of the limitations, the obtained results are just slightly worse than the best literature results. Besides, fixed temperature intervals make the model easier to solve, especially for more complex problems, such as large-scale problems.

This example was further extended by Ibrić et al.,³⁸ involving two freshwater sources (20, 80 °C) and three WTs. Besides, the temperature of discharged water is assumed to be 30 °C. The optimized result is obtained for $\Delta T_{\min} = 10^\circ\text{C}$, as shown in Figure 10. The TAC of the obtained network is 1915940.6 \$/y, with the freshwater consumption of 40.466 kg/s (20 °C, 40.000 kg/s; 80 °C, 0.466 kg/s). Compared to the result from Ibrić et al.³⁸ (recalculated according to the published result), the freshwater cost increases 33.5%, as shown in Table 4. However, the operating cost and investment cost of the wastewater treatment units decrease, since more freshwater are consumed. Besides, better heat integration opportunities are found which lead to both the smaller investment cost for heat exchangers and the minor utility cost. In Ibrić et al.,³⁸ a two-step solution strategy was adopted to solve the HIWN problem. The second step was solved to minimize the TAC with the constraints of freshwater flow rate, wastewater flow rate, and utility consumption from the first step. Nevertheless, only the operating cost was minimized in the first step. The trade-offs among the heat integration, the freshwater consumption, and the wastewater treatment were not fully considered in the first step. Although a network with less operating cost was found in Ibrić et al.,³⁸ it led to higher investment cost of wastewater treatment units and heat exchangers. The optimized result in this work is better than the literature result,³⁸ with 0.47% decrease of TAC.

■ RESULT AND DISCUSSION

The results and computational time of each step for examples 1–4 are given in Table 5. Although the rTAC is optimized in the second step, the accurate TAC can be calculated, namely cTAC, according to the obtained results. It can be seen that the error between the rTAC and cTAC is smaller than 10%. Besides, the

cTAC is close to the TAC obtained in the final step, which means that acceptable results can be obtained by minimizing rTAC. In order to demonstrate the advantage of the proposed solution strategy, four examples are solved by another solution strategy (strategy OC), where the operating cost (OC) is optimized in the second step. The results (Table 6) indicate that the obtained results by strategy OC are not better than the ones in Table 5, except example 1. What is more, the cTAC is much worse than the ones in Table 5. In summary, the solution strategy minimizing rTAC in the second step performs better than the one minimizing OC. Because the trade-offs between the operating cost and the underestimated capital cost are considered during minimizing rTAC, better initial points can be generated, which will lead to better results in the last step.

CONCLUSION

From a sustainable perspective, multiple contaminant constraints, wastewater treatment as well as energy consumption should all be involved in the design problem of HIWN. However, due to the nonconvex and nonlinear nature of this problem, it is difficult to solve especially for the large-scale cases. To address this difficulty of large-scale cases, a novel model and a solution strategy are proposed in this paper. The proposed HIWN model originates from our newly developed TAC targeting approach⁴³ and HEN model.⁴⁴ These models are extended to involve the multiple contaminants constraints and wastewater treatment units. Besides, instead of targeting the simplified TAC, the accurate TAC is optimized in this paper. The proposed model allows nonisothermal mixing while eliminates the nonlinear terms in the HEN constraints. Consequently, the proposed model contains more structure possibilities. Nevertheless, the nonlinear terms still exist in the WN constraints, solving the model is still challenging. Thus, a novel three-step solution strategy is proposed to obtain promising solutions. Four literature examples, including two large-scale examples, are illustrated to demonstrate the applicability of the proposed model and the efficiency of the solution strategy. Best results are obtained for all examples, except the first case of example 4, whose result is very close to the best reported one. Significantly, the feasibility of the proposed model and the solution strategy is highlighted when the complexities of problems increase, such as involving more freshwater sources, wastewater treatment units, and water-using units. Future research can be directed toward the planning of eco-industrial parks.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acssuschemeng.7b03740.

Formulations for water-using and wastewater treatment network, mass flow pattern, heat flow pattern, and identification of heat exchangers; data of water-using and wastewater treatment units for examples 1–4; operating parameter and cost data for examples 1–4 (PDF)

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Notes

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ACKNOWLEDGMENTS

The financial support provided by the Project of National Natural Science Foundation of China (91434205 and 61590925), the National Science Fund for Distinguished Young (21525627), and the International S&T Cooperation Projects of China (2015DFA40660) are gratefully acknowledged.

NOTATION

Subscript

i	source (include outlet of each unit and freshwater)
j	demand (include inlet of each unit and wastewater)
k	temperature interval
l	hot stream
l_s	hot stream substream
l_s	hot source stream
l_d	hot demand stream
m	cold stream
m_s	cold stream substream
m_s	cold source stream
m_d	cold demand stream
n	temperature level

Sets

C	cold stream, $C = (m m = 1, 2, 3, \dots, 2N_s)$, $C = \text{CDUCS}$
CD	cold demand stream, $CD = (m_d m_d \in D, m_d = N_s + 1, N_s + 2, \dots, 2N_s)$
C_k	cold stream in temperature interval k , $C_k \in C$
CMS	cold substream, $CLS = ((m, ms) m \in C, ms \in MS, (m, ms) = (1, 1), (2, 1), \dots)$
CS	cold source stream, $CS = (m_s m_s \in D, m_s = 1, 2, 3, \dots, N_s)$
D	demand, $D = (j j = N_s + 1, N_s + 2, \dots, 2N_s(N_D))$, $D = \text{DWUWUDWTD}$
DW	discharged wastewater, $DW \subset D$
FW	freshwater—water source, $FW \subset C$
H	hot stream, $H = (l l = 1, 2, 3, \dots, 2N_s)$, $H = \text{HDUHS}$
HD	hot demand stream, $HD = (l_d l_d \in H, l_d = N_s + 1, N_s + 2, \dots, 2N_s)$
H_k	hot stream in temperature interval k , $H_k \subset H$
HLS	hot substream, $HLS = ((l, l_s) l \in H, l_s \in LS, (l, l_s) = (1, 1), (2, 1), \dots)$
HS	hot source stream, $HS = (l_s l_s \in H, l_s = 1, 2, 3, \dots, N_s)$
LS	substream for hot stream, $LS = (l_s l_s = 1, 2, 3, \dots)$
MS	substream for cold stream, $MS = (m_s m_s = 1, 2, 3, \dots)$
S	source, $S = (i i = 1, 2, 3, \dots, N_s)$, $S = \text{FWUWUSUWTS}$
TL	temperature level, $TL = (n n = 1, 2, 3, \dots, N_K + 1)$
TI	temperature interval, $TI = (k k = 1, 2, 3, \dots, N_K)$
WUD	inlet of water-using unit, $WUD \subset D$
WUS	outlet of water-using unit, $WUS \subset C$
WTD	inlet of wastewater treatment unit, $WTD \subset D$
WTS	outlet of wastewater treatment unit, $WTS \subset C$

Variables

$c_{jin,ic}$	inlet concentration of each demand, ppm
$c_{iout,ic}$	outlet concentration of each source, ppm
d_{tcu_l}	temperature difference of cooler between hot stream l and cold utility, °C
d_{thu_m}	temperature difference of heater between cold stream m and hot utility, °C

$dt_{l,ls,m,ms,k}$	temperature difference of heat exchanger between hot stream substream l,ls and cold stream substream m,ms in temperature interval $k,^{\circ}\text{C}$
FWC	freshwater cost, $\$/\text{kg}$
f_i	outlet flow rate of each unit, kg/s
f_j	inlet flow rate of each unit, kg/s
fl_i	flow rate of hot demand stream, kg/s
$flk_{i,k}$	flow rate of hot demand stream in temperature interval k , kg/s
fl_{ls}	flow rate of hot source stream, kg/s
$flk_{ls,k}$	flow rate of hot source stream in temperature interval k , kg/s
$flsk_{ls,k}$	flow rate of hot substream in temperature interval k , kg/s
$fm_{m,i}$	flow rate of cold demand stream, kg/s
$fmk_{m,i,k}$	flow rate of cold demand stream in temperature interval k , kg/s
$fm_{m,i}$	flow rate of cold source stream, kg/s
$fmk_{m,i,k}$	flow rate of cold source stream in temperature interval k , kg/s
$fmsk_{m,ms,k}$	flow rate of cold substream in temperature interval k , kg/s
$fresh$	freshwater consumption, kg/s
$hcd_{m,i}$	residual energy of cold demand stream, kW
$hhd_{l,i}$	residual energy of hot demand stream, kW
qc_l	heat load of cooler for hot stream l , kW
qh_m	heat load of heater for cold stream m , kW
$q_{l,ls,m,ms,k}$	heat load of heat exchanger between hot substream l,ls and cold substream m,ms in temperature interval k , kW
$rqhs_{l,ls,n}$	residual energy of hot substream l,ls in temperature level n
$rqcs_{m,ms,n}$	residual energy of cold substream m,ms in temperature level n
$rTAC$	relaxed total annual cost, $\$$
sd_{ij}	flow rate from source i to demand j , kg/s
TAC	total annual cost, $\$$
$tic_{m,ms,k}$	inlet temperature of cold substream m,ms in temperature interval $k,^{\circ}\text{C}$
$tih_{l,ls,k}$	inlet temperature of hot substream l,ls in temperature interval $k,^{\circ}\text{C}$
$toc_{m,ms,k}$	outlet temperature of cold substream m,ms in temperature interval $k,^{\circ}\text{C}$
$toh_{l,ls,k}$	outlet temperature of hot substream l,ls in temperature interval $k,^{\circ}\text{C}$
$trh_{l,l,i,n}$	mass flow between hot source stream and hot demand stream, kg/s
$trc_{m,m,i,n}$	mass flow between cold source stream and cold demand stream, kg/s

Binary variables

$mlsk_{l,ls,k}$	monotonicity of flow rate for hot substream
$mmsk_{m,ms,k}$	monotonicity of flow rate for cold substream
zc_l	existence of the cooler for hot stream l
zh_m	existence of the heater for cold stream m
$z_{l,ls,m,ms,k}$	existence of the heat transfer match between hot substream l,ls and cold substream m,ms in temperature interval k
$zlsk_{l,ls,k}$	consistency of flow rate for hot substream l,ls
$zmsk_{m,ms,k}$	consistency of flow rate for cold substream m,ms
$zz_{l,ls,m,ms,k}$	existence of the heat exchanger between hot substream l,ls and cold substream m,ms in temperature interval k

Parameters

α_i	cost exponent for wastewater treatment unit
AF	annualized investment factor for wastewater treatment unit
$B_{l,m}$	area cost exponent for heat exchanger
$B_{l,CU}$	area cost exponent for cooler
$B_{m,HU}$	area cost exponent for heater
$C_{l,m}$	annualized area cost coefficient for heat exchanger, $\$/\text{m}^2$
$C_{l,CU}$	annualized area cost coefficient for cooler, $\$/\text{m}^2$
$C_{m,HU}$	annualized area cost coefficient for heater, $\$/\text{m}^2$
CCU	per unit cost for cold utility, $\$/(\text{kW}\cdot\text{y})$
$CF_{l,m}$	fixed charge for exchangers, $\$$
$CF_{l,CU}$	fixed charge for cooler, $\$$
$CF_{m,HU}$	fixed charge for heater, $\$$
CFW_i	cost of freshwater, $\$/\text{kg}$
CHU	per unit cost for hot utility, $\$/(\text{kW}\cdot\text{y})$
CT_i	cost coefficient for wastewater treatment unit
CWT_i	operating cost for wastewater treatment unit, $\$/\text{kg}$
$c_{jinm_{i,c}}$	limited inlet concentration of each demand, ppm
$cioutm_{i,c}$	limited outlet concentration of each source, ppm
c_p	heat capacity of water, $\text{kJ}/(\text{kg}\cdot^{\circ}\text{C})$
H	hours of plant operation per annum, h
hot_{ij}	identity of mass flow sd_{ij}
$load_{i,c}$	mass load of water-using unit, g/s
$MFLn_{l,n}$	maximum residual energy for hot stream l in temperature level n , kW
$MFmn_{m,n}$	maximum residual energy for cold stream m in temperature level n , kW
ne_l	ending temperature level for hot stream l , $ne_l \in \text{TL}$
ne_m	starting temperature level for cold stream m , $ne_m \in \text{TL}$
ns_l	starting temperature level for hot stream l , $ns_l \in \text{TL}$
ns_m	starting temperature level for cold stream m , $ns_m \in \text{TL}$
$rdtcu_l$	relaxed temperature difference of cooler, $^{\circ}\text{C}$
$rdthu_m$	relaxed temperature difference of heater, $^{\circ}\text{C}$
$rdt_{l,ls,m,ms,k}$	relaxed temperature difference of heat exchanger, $^{\circ}\text{C}$
$rr_{i,c}$	removal ratio of contaminant c in wastewater treatment unit i
tce_m	ending temperature of cold stream $m,^{\circ}\text{C}$
tcs_m	starting temperature of cold stream $m,^{\circ}\text{C}$
the_l	ending temperature of hot stream $l,^{\circ}\text{C}$
ths_l	starting temperature of hot stream $l,^{\circ}\text{C}$
tn_n	temperature for temperature level $n,^{\circ}\text{C}$
$U_{l,m}$	heat transfer coefficient between hot and cold streams, $\text{kW}/(\text{m}^2\cdot^{\circ}\text{C})$
$U_{l,CU}$	heat transfer coefficient between hot stream and cold utility, $\text{kW}/(\text{m}^2\cdot^{\circ}\text{C})$
$U_{m,HU}$	heat transfer coefficient between cold stream and hot utility, $\text{kW}/(\text{m}^2\cdot^{\circ}\text{C})$

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